

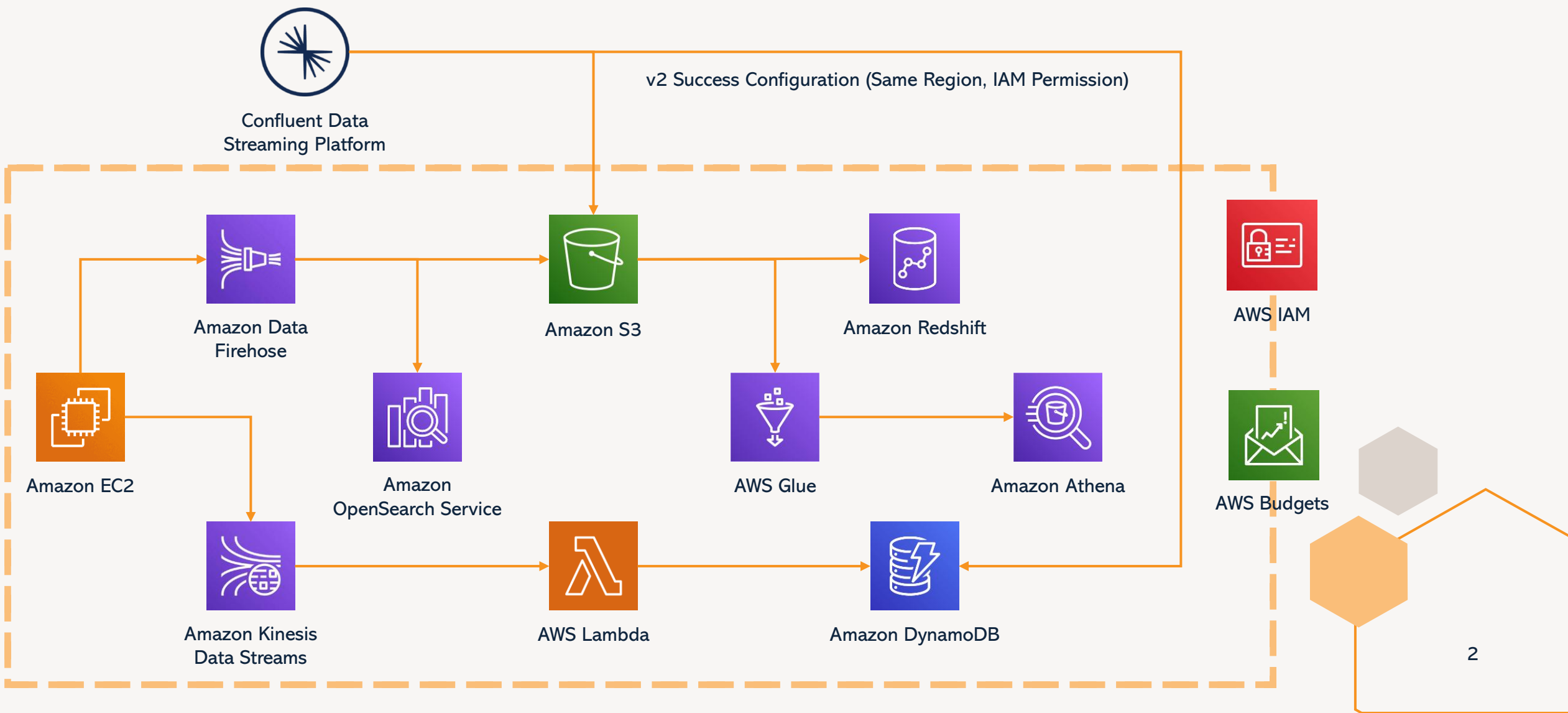
Modern Data Platform Using Confluent and AWS

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April 17, 2024



Data Movement Diagram



Actual Cost (My Account, Free Tier \$100)

The screenshot displays the AWS Billing and Cost Management console. The left sidebar contains navigation links for Billing and Cost Management, Home, Getting Started, Billing and Payments, Bills, Payments, Credits, Purchase Orders, Cost Analysis, Cost Explorer, Cost Explorer Saved Reports, Cost Anomaly Detection, and Free Tier. The main content area shows the 'Bills' page for the billing period of April 2024. The 'AWS estimated bill summary' section provides details on the account ID, billing period (April 1 - April 30, 2024), and bill status (Pending). A table lists the service provider as Amazon Web Services, Inc. with a total cost of USD 0.88. The estimated grand total is also shown as USD 0.88.

Account ID	Billing period	Bill status
[REDACTED]	April 1 - April 30, 2024	Pending

Service provider	Total in USD
Amazon Web Services, Inc.	USD 0.88

Estimated grand total: USD 0.88

Confluent Cloud (Free \$400 Credit)

The screenshot displays the Confluent Cloud user interface. The top navigation bar includes the Confluent logo, a search bar for 'Stream Catalog', a 'LEARN' button, and notification and help icons. The left sidebar contains navigation links for Home, Environments (highlighted), Data portal, Stream processing (marked as 'New'), Cluster links, Stream shares, and Support. The main content area is titled 'Live (1)' and features a cluster card for 'test-conf-cluster' in a 'Running' state. Below the cluster card, a 'Metrics' section shows Production at 0B/s, Consumption at 0B/s, and Storage at 13.24KB. A 'Resources' section shows 1 ksqIDB, 0 Connectors, and 23 Clients. An 'Overview' section is partially visible at the bottom. The right sidebar shows options to add tags, show more resources, and a 'Stream Governance Essentials' section with an 'Upgrade now' button.

CONFLUENT Stream Catalog LEARN

Home Environments Data portal Stream processing **New** Cluster links Stream shares Support

Live (1)

 **test-conf-cluster** Running ...

Metrics

Production	Consumption	Storage
0B/s	0B/s	13.24KB

Resources

ksqIDB	Connectors	Clients
1	0	23

Overview

Add tags to this environment

▼ Show more resources

Stream Governance Essentials

Amazon Web Services

Upgrade now

2

Schedule

Support

Confluent Cloud Topics

CONFLUENT

Stream Catalog

LEARN

Search topics

Hide internal topics

+ Add topic

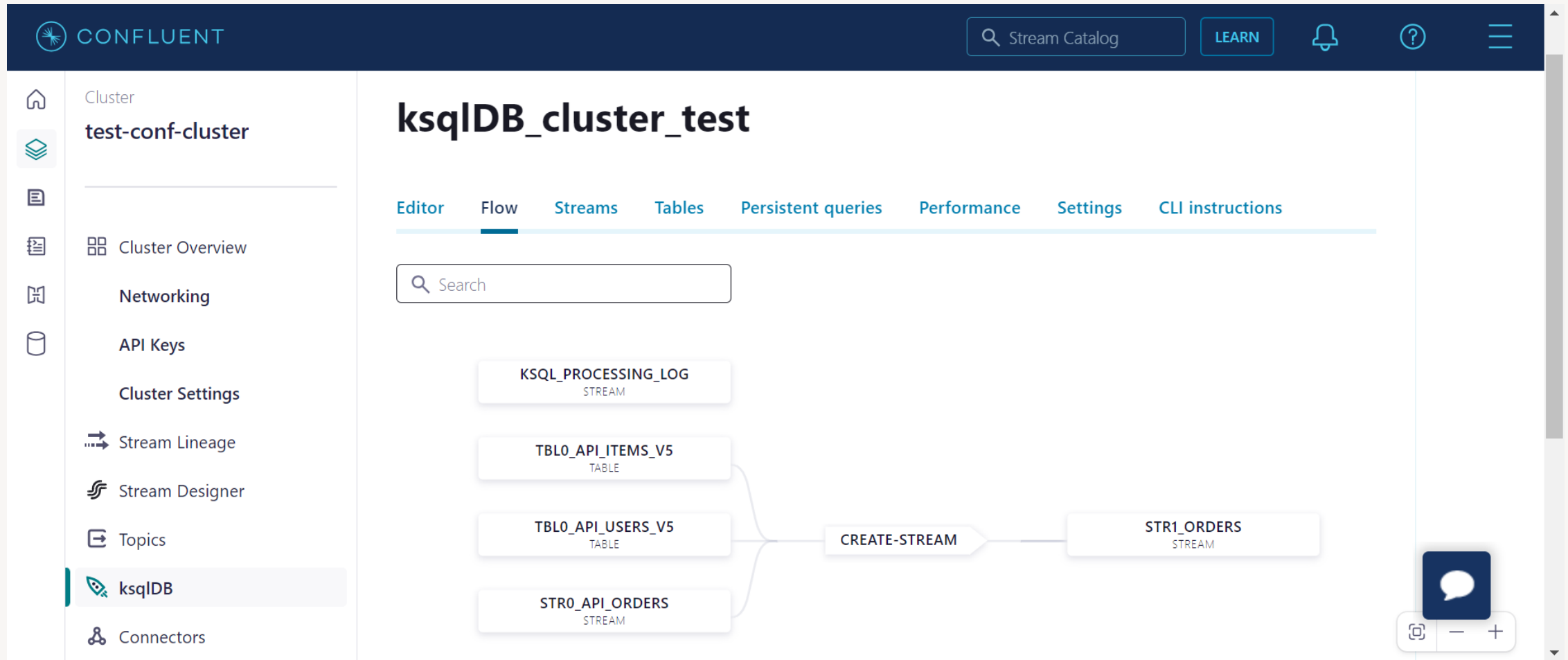
Topic name	Tags	Partitions	Production	Consumption
dlq-lcc-56xmdz	--	1	--	--
pksqlc-yoxmpp-processing-log	--	8	0B/s	0B/s
sink t1 orders	--	1	0B/s	0B/s
src-api-items	--	1	0B/s	0B/s
src-api-orders	--	1	--	0B/s
src-api-users	--	1	--	0B/s

ksqlDB Cluster

The screenshot shows the Confluent web interface for managing a ksqlDB cluster. The top navigation bar includes the Confluent logo, a search bar for the Stream Catalog, a 'LEARN' button, and notification and help icons. The breadcrumb trail indicates the path: Home > Environments > default > test-conf-cluster >. The left sidebar contains a 'Cluster' section with 'test-conf-cluster' selected, and a list of navigation items: Cluster Overview, Networking, API Keys, Cluster Settings, Stream Lineage, Stream Designer, Topics, ksqlDB (highlighted), and Connectors. The main content area is titled 'ksqlDB' and features a search bar and an '+ Add cluster' button. Below these is a table listing the cluster details.

Cluster name	Status	Actions	Streaming units	Allowec
<u>ksqlDB cluster test</u>	Up	Delete	1	40

ksqlDB Data Flow



ksqlDB Stream List

CONFLUENT

Stream Catalog

LEARN

Home / Environments / default / test-conf-cluster / ksqlDB

Cluster

test-conf-cluster

Cluster Overview

Networking

API Keys

Cluster Settings

Stream Lineage

Stream Designer

Topics

ksqlDB

Connectors

ksqlDB_cluster_test

EditorFlowStreamsTablesPersistent queriesPerformanceSettingsCLI instructions

Search streams

+ Import topics as streams

Stream	Kafka topic			Data
Name	Topic Name	Partitions	Replication	Key Format
<u>STR1_ORDERS</u>	sink_t1_orders	1	3	AVRO
<u>KSQL_PROCESSING_LOG</u>	pksqlc-yoxmpp-p...	8	3	KAFKA
<u>STR0_API_ORDERS</u>	src-api-orders	1	3	KAFKA

ksqlDB Table List

 CONFLUENT

Stream Catalog

LEARN







Home > Environments > default > test-conf-cluster > ksqlDB >

 Cluster

 test-conf-cluster

 Cluster Overview

 Networking

 API Keys

 Cluster Settings

 Stream Lineage

 Stream Designer

 Topics

 ksqlDB

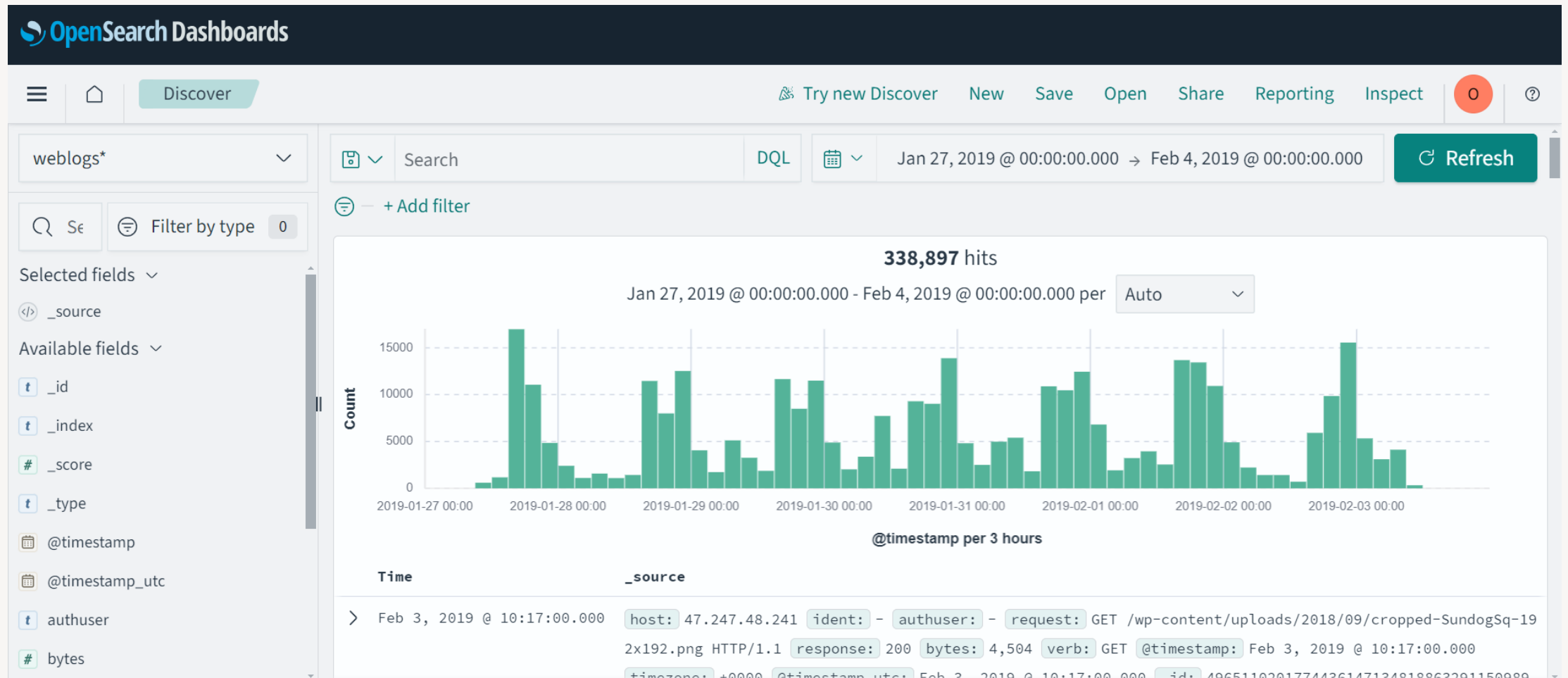
ksqlDB_cluster_test

EditorFlowStreams**Tables**Persistent queriesPerformanceSettingsCLI instructions

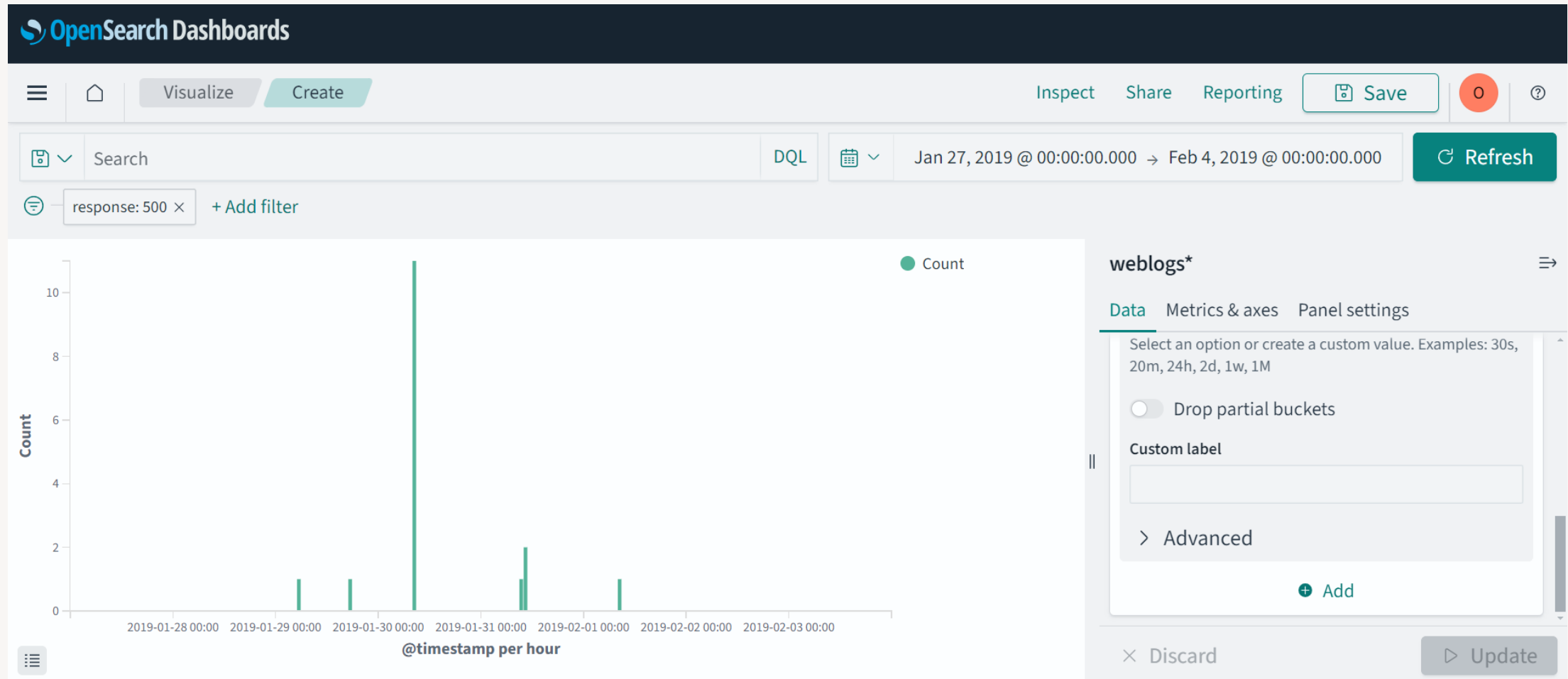
Search tables

Table	Kafka topic			Data	
	Name	Topic Name	Partitions	Replication	Key Format
<u>TBL0_API_USERS_V5</u>		src-api-users	1	3	KAFKA
<u>TBL0_API_ITEMS_V5</u>		src-api-items	1	3	KAFKA

Amazon OpenSearch Service



Amazon OpenSearch Service



OpenSearch Index

The screenshot shows the OpenSearch Dashboards interface for the 'weblogs*' index pattern. The left sidebar contains 'Dashboards Management' with links to 'Index patterns', 'Saved objects', and 'Advanced settings'. The main content area displays the index name 'weblogs*' and a time field '@timestamp'. Below this, a message states: 'This page lists every field in the **weblogs*** index and the field's associated core type as recorded by OpenSearch. To change a field type, use the OpenSearch [Mapping API](#).' The 'Fields (21)' tab is selected, showing a search bar and a dropdown for 'All field types'. A table lists the fields:

Name	Type	Format	Searchable	Aggregatable	Excluded
@timestamp ⓘ	date		●	●	✎
@timestamp_utc	date		●	●	✎
id	string		●	●	✎

Amazon OpenSearch Service Cluster Domain

The screenshot displays the Amazon OpenSearch Service Dashboard. The left sidebar contains navigation links for 'Managed clusters' (Dashboard, Domains, Reserved Instance leases, Packages, VPC endpoints) and 'Serverless' (Dashboard, Collections, Security, SAML authentication, Data access policies). The main content area is titled 'Dashboard' and features a 'Domain health' section with a summary of domain status: 1 All domains, 0 Red, 1 Yellow, and 0 Green. Below this is a table of domains with columns: Name, Domai..., Change..., Engine, Version, Cluster..., and Total fr... The table lists one domain, 'testopensearch', which is 'Active' and 'Completed', using the 'OpenSearch' engine, version '2.11', with a 'Yellow' cluster status and a total size of '7.69 GiB'. A 'View all domains' link is at the bottom of the table. The top navigation bar includes the AWS logo, 'Services', a search bar, and regional/user information (Singapore, Yunata). The bottom footer contains 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon OpenSearch Service

Managed clusters

- Dashboard
- Domains
- Reserved Instance leases
- Packages
- VPC endpoints

Serverless

- Dashboard
- Collections [New](#)
- Security
- SAML authentication
- Data access policies

Dashboard

Order Reserved Instances

Create domain

Domain health [Info](#)

All domains

1

0

1

0

Red

Yellow

Green

Name	Domai...	Change...	Engine	Version	Cluster...	Total fr...
testopensearch	Active	Completed	OpenSearch	2.11	Yellow	7.69 GiB

[View all domains](#)

CloudShell Feedback

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Amazon OpenSearch Service

The screenshot displays the Amazon OpenSearch Service console interface. The left sidebar contains navigation links for 'Managed clusters' (Dashboard, Domains, Reserved Instance leases, Packages, VPC endpoints) and 'Serverless' (Dashboard, Collections, Security, SAML authentication, Data access policies). The main content area shows the 'testopensearch' domain details under the 'Domains' section. The breadcrumb trail is 'Amazon OpenSearch Service > Domains > testopensearch'. The domain name 'testopensearch' is prominently displayed with an 'Info' link. To the right of the domain name are 'Delete' and 'Actions' buttons. Below this is a 'General information' section containing a table of domain details.

General information			
Name testopensearch	Domain processing status Info ✓ Active	Version Info OpenSearch 2.11 (latest)	OpenSearch Dashboards URL (dual stack) https://search-testopensearch-k5i6ogxlctmoob3xumwne3cofa.aos.ap-southeast-1.on.aws/_dashboards
Domain ARN arn:aws:es:ap-southeast-1:654654377830:domain/testopensearch	Configuration change status Info ✓ Completed	Service software version Info OpenSearch_2_11_R20231113-P4 (latest)	IPv4 URL
	Cluster health Info Yellow		Domain endpoint v2 (dual stack)

The footer of the console includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon Redshift (Query Editor V2)

The screenshot displays the Amazon Redshift Query Editor V2 interface. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and a keyboard shortcut '[Alt+S]'. On the right, there are icons for a document, a bell, a help icon, a settings icon, and location/user dropdowns for 'Singapore' and 'Yunata'.

The left sidebar contains a navigation pane with sections for 'Editor', 'Queries', and 'Notebooks'. The 'Queries' section is expanded, showing a tree view of resources under 'test-redshift' and 'dev'. The 'dev' folder contains an 'orderlog_schema' folder, which has a 'Tables' sub-folder with two tables: 'es_error' and 'year_20...'. There is also a 'public' folder and a 'sample_data_dev' folder.

The main editor area is titled 'Redshift query editor v2' and shows a tab for 'Untitled 1'. It includes a 'Run' button, a 'Limit 100' toggle, an 'Explain' toggle, an 'Isolated session' toggle, and dropdowns for 'test-redshift' and 'dev'. Below these are 'Schedule', 'Save', and 'Fullscreen' icons. The SQL query is as follows:

```
1 CREATE EXTERNAL SCHEMA orderlog_schema FROM data catalog
2 database 'orderlogs'
3 iam_role 'arn:aws:iam::654654377830:role/Redshift-Spectrum'
4 region 'ap-southeast-1';
5
6 SELECT description, COUNT(1) AS order_count FROM orderlog_schema.year_2024
7 WHERE country = 'France' GROUP BY description;
```

The results section shows 'Result 1 (100)' with an 'Export' button and a 'Chart' toggle. The results are displayed in a table:

	description	order_count
☐	SET OF 5 LUCKY CAT M...	10
☐	ICE CREAM BUBBLES	16
☐	RED RETROSPOT MINI ...	52
☐	BLACK RECORD COVE...	6

At the bottom of the results section, it shows 'Query ID 1747', 'Elapsed time: 2664 ms', and 'Total rows: 100'. The footer of the interface includes 'CloudShell', 'Feedback', and copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates.', along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon Redshift Cluster

The screenshot displays the Amazon Redshift Provisioned clusters dashboard. At the top, there's a navigation bar with the AWS logo, 'Services' link, a search bar, and utility icons. The main header shows 'Amazon Redshift > Provisioned clusters dashboard'. Below this, the title 'Provisioned clusters dashboard' is followed by an 'Info' link and three buttons: 'Try Amazon Redshift Serverless', 'Purchase reserved nodes', and a prominent orange 'Create cluster' button.

The 'Resources overview' section provides resource data for the Asia Pacific (Singapore) Region. It includes a table with the following data:

Total nodes	On-demand nodes	Reserved nodes	Reserved nodes available	Automated snapshots	Manual snapshots
1	1	0	0 (0 of 0 used)	0	0

Below the overview, the 'Cluster overview (1)' section features a dropdown menu set to 'Any status'. It contains a table with one cluster:

Cluster	Status
test-redshift	Available

To the right of the cluster overview is a 'Datashares' section with a 'Require authorization' warning icon and text explaining how to authorize other AWS accounts.

The footer includes links for 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon Redshift Cluster Detail

The screenshot displays the Amazon Redshift console interface. At the top, the AWS logo and 'Services' menu are visible, along with a search bar and navigation icons. The breadcrumb trail indicates the path: Amazon Redshift > Clusters > test-redshift. The cluster name 'test-redshift' is prominently displayed, followed by action buttons: 'Actions', 'Edit', 'Add partner integration', and 'Query data'. Below this, the 'General information' tab is selected, showing a table of cluster details. The table includes fields such as Cluster identifier, Status (Available), Node type (dc2.large), Endpoint, Custom domain name, Date created, Number of nodes, Cluster namespace ARN, Storage used, Multi-AZ, Cluster configuration, and Production status. The footer contains links for CloudShell, Feedback, and copyright information.

General information Info			
Cluster identifier test-redshift	Status ✓ Available	Node type dc2.large	Endpoint test-redshift.ckymznaeiuy.ap-southeast-1.redshift.amazonaws.com:5439/dev
Custom domain name -	Date created April 14, 2024, 13:56 (UTC+07:00)	Number of nodes 1	JDBC URL jdbc:redshift://test-redshift.ckymznaeiuy.ap-southeast-1.redshift.amazonaws.com:5439/dev
Cluster namespace ARN arn:aws:redshift:ap-southeast-1:654654377830:namespace:13754933-49ca-48ac-8977-a7c89fe2da48	Storage used 0.02% (0.03 of 160 GB used)		ODBC URL Driver={Amazon Redshift (x64)}; Server=test-redshift.ckymznaeiuy.ap-southeast-1.redshift.amazonaws.com:5439/dev
Cluster configuration Production	Multi-AZ No		

Amazon EC2

The screenshot displays the Amazon EC2 console interface. The top navigation bar includes the AWS logo, a search bar, and regional settings for Singapore and Yunata. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, and a dropdown menu for Instances, which is currently expanded to show options like Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity, and Reservations. The main content area shows the 'Instances (1/1)' page with a search bar and a filter set to 'Instance state = running'. A table lists the instance 'Order Data So...' with ID 'i-013a60e2861c7faf7', state 'Running', type 't2.micro', and status '2/2 checks passed'. Below the table, a detailed view for the selected instance shows its configuration: Instance ID 'i-013a60e2861c7faf7 (Order Data Source)', Public IPv4 address '52.221.239.114', Private IPv4 addresses '172.31.29.135', Instance state 'Running', Public IPv4 DNS 'ec2-52-221-239-114.ap-southeast-1.compute.amazonaws.com', Private IP DNS name 'ip-172-31-29-135.ap-southeast-1.compute.amazonaws.com', and Hostname type 'IP name: ip-172-31-29-135.ap-southeast-1.compute.amazonaws.com'. The bottom of the console features a CloudShell button, a feedback link, and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for Privacy, Terms, and Cookie preferences.

aws Services Search [Alt+S] Singapore Yunata

EC2 Dashboard X

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity

Reservations [New](#)

▼ Images

AMIs

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters

Connect Instance state Actions Launch instances

All states

< 1 > ⚙

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☒	Order Data So...	i-013a60e2861c7faf7	Running	t2.micro	2/2 checks passed	View alarms +	ap-southeast-1

Instance: i-013a60e2861c7faf7 (Order Data Source)

Instance ID
i-013a60e2861c7faf7 (Order Data Source)

Public IPv4 address
52.221.239.114 [open address](#)

Private IPv4 addresses
172.31.29.135

IPv6 address
-

Instance state
Running

Public IPv4 DNS
ec2-52-221-239-114.ap-southeast-1.compute.amazonaws.com [open address](#)

Hostname type
IP name: ip-172-31-29-135.ap-southeast-1.compute.amazonaws.com

Private IP DNS name (IPv4 only)
ip-172-31-29-135.ap-southeast-1.compute.amazonaws.com

CloudShell Feedback

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Amazon EC2 Kinesis Agent

```
ec2-user@ip-172-31-29-135:~  
GNU nano 5.8 /etc/aws-kinesis/agent.json  
{  
  "cloudwatch.emitMetrics": true,  
  "kinesis.endpoint": "",  
  "firehose.endpoint": "firehose.ap-southeast-1.amazonaws.com",  
  "kinesis.endpoint": "kinesis.ap-southeast-1.amazonaws.com",  
  
  "flows": [  
    {  
      "filePattern": "/var/log/httpd/ssl_access*",  
      "deliveryStream": "WebLogsStream",  
      "initialPosition": "START_OF_FILE"  
    }  
  ]  
}
```

[Read 14 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

Amazon EC2 Kinesis Agent

```
ec2-user@ip-172-31-29-135:~
GNU nano 5.8 /etc/aws-kinesis/agent-1.json
{
  "cloudwatch.emitMetrics": true,
  "kinesis.endpoint": "",
  "firehose.endpoint": "firehose.ap-southeast-1.amazonaws.com",
  "kinesis.endpoint": "kinesis.ap-southeast-1.amazonaws.com",

  "flows": [
    {
      "filePattern": "/var/log/test/*.log",
      "kinesisStream": "Test-Orders",
      "partitionKeyOption": "RANDOM",
      "dataProcessingOptions": [
        {
          "optionName": "CSVTOJSON",
          "customFieldNames": ["InvoiceNo", "StockCode", "Description", "Quan>
      ]
    },
    {
      "filePattern": "/var/log/test/*.log",
      "deliveryStream": "PUT-S3-0sgff"
    }
  ]
}

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

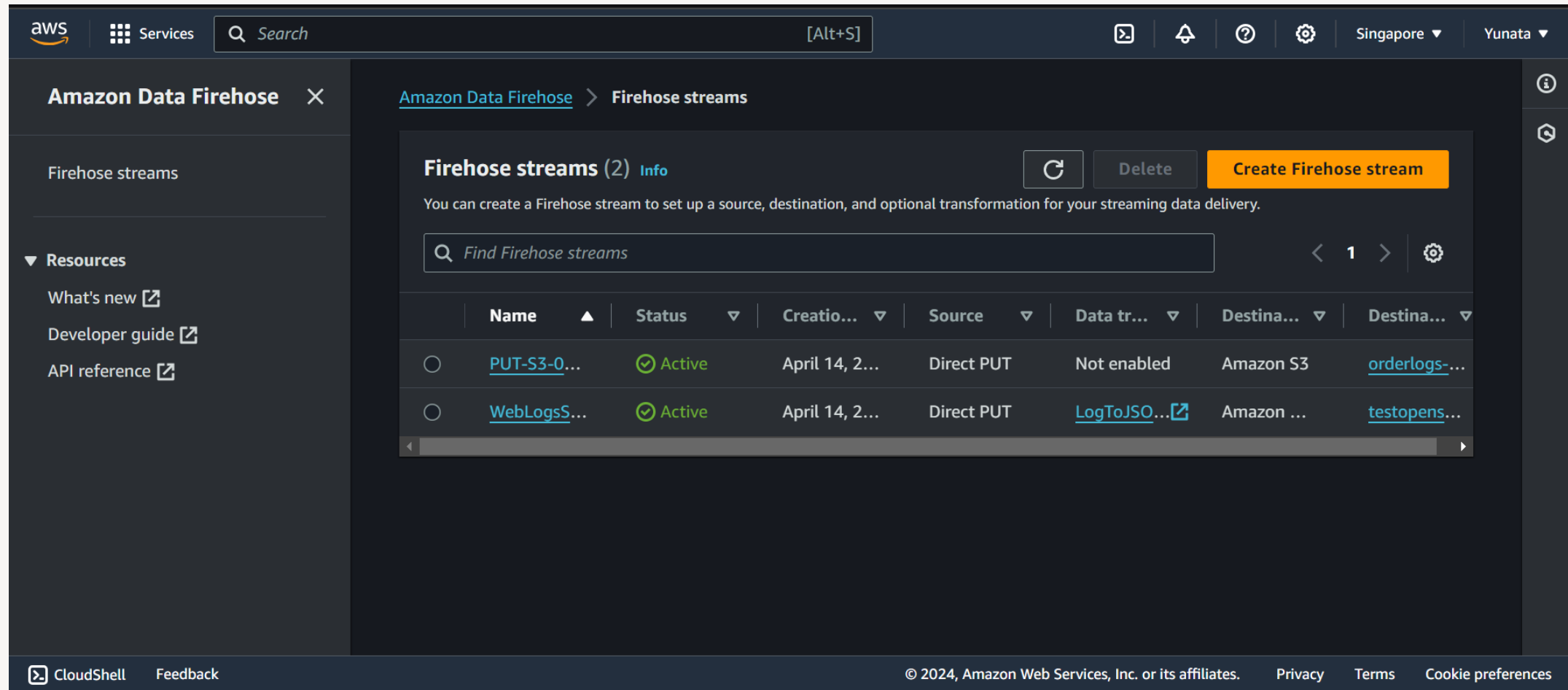
Kinesis Data Stream

The screenshot displays the Amazon Kinesis console interface. The top navigation bar includes the AWS logo, 'Services' menu, a search bar, and user information for 'Singapore' and 'Yunata'. The left sidebar shows the 'Amazon Kinesis' section with a 'Dashboard' link and a 'Data streams' link. Under 'Data streams', there are links for 'Amazon Data Firehose' and 'Managed Apache Flink', both marked as 'New'. The main content area shows the 'Test-Orders' data stream details. The breadcrumb navigation is 'Amazon Kinesis > Data streams > Test-Orders'. The title 'Test-Orders' is followed by an 'Info' link and a 'Delete' button. Below this is a 'Data stream summary' section with a table of key attributes:

Status	Capacity mode	ARN	Creation time
Active	On-demand	arn:aws:kinesis:ap-southeast-1:654654377830:stream/Test-Orders	April 14, 2024 at 10:03 GMT+7
	Data retention period		
	1 day		

Below the summary is a navigation bar with tabs: 'Applications', 'Monitoring' (selected), 'Configuration', 'Enhanced fan-out (0)', 'Data viewer', and 'Data analytics'. The 'Monitoring' tab is active, showing the 'Stream metrics' section with an 'Info' link. The footer of the console includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Data Firehose (Former: Kinesis Data Firehose)



The screenshot displays the Amazon Data Firehose console interface. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and a keyboard shortcut '[Alt+S]'. On the right side of the top bar, there are icons for CloudShell, notifications, help, settings, and regional selection (Singapore and Yunata).

The left sidebar contains the 'Amazon Data Firehose' title with a close button, a 'Firehose streams' link, and a 'Resources' section with links for 'What's new', 'Developer guide', and 'API reference'.

The main content area is titled 'Amazon Data Firehose > Firehose streams'. It features a 'Firehose streams (2) Info' header with a refresh button, a 'Delete' button, and a prominent orange 'Create Firehose stream' button. Below the header, a descriptive text states: 'You can create a Firehose stream to set up a source, destination, and optional transformation for your streaming data delivery.' A search bar labeled 'Find Firehose streams' is positioned above a table of streams.

	Name	Status	Creatio...	Source	Data tr...	Destina...	Destina...
☐	PUT-S3-0...	Active	April 14, 2...	Direct PUT	Not enabled	Amazon S3	orderlogs-...
☐	WebLogsS...	Active	April 14, 2...	Direct PUT	LogToJSO...	Amazon ...	testopens...

The bottom of the console shows a 'CloudShell' button, a 'Feedback' link, and a footer with copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates.' along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Data Firehose Stream

The screenshot displays the Amazon Data Firehose console interface. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and utility icons for CloudShell, notifications, help, settings, and regional/user selection (Singapore, Yunata). The left sidebar shows the 'Amazon Data Firehose' title and a 'Firehose streams' section. Under 'Resources', there are links for 'What's new', 'Developer guide', and 'API reference'. The main content area shows the breadcrumb path 'Amazon Data Firehose > Firehose streams > PUT-S3-0sgff'. The stream name 'PUT-S3-0sgff' is prominently displayed with an 'Info' link and a 'Delete Firehose stream' button. Below this, the 'Firehose stream details' section is organized into a grid:

Status ✔ Active	Destination Amazon S3	Data transformation Not enabled	Creation time April 14, 2024 at 09:17 GMT+7
Source Direct PUT	ARN arn:aws:firehose:ap-southeast-1:654654377830:deliverystream/PUT-S3-0sgff	Dynamic partitioning Not enabled	Error logs status ✔ 0 Destination error logs

At the bottom of the details section, there is a 'Test with demo data' button with an 'Info' link. A note below it states: 'Ingest simulated data to test the configuration of your Firehose stream. Standard Amazon Data Firehose charges apply.' The footer contains links for 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Data Firehose Stream

The screenshot displays the Amazon Data Firehose console interface. At the top, the AWS logo and 'Services' menu are visible, along with a search bar and navigation icons. The breadcrumb trail indicates the path: [Amazon Data Firehose](#) > [Firehose streams](#) > [WebLogsStream](#). The main heading is 'WebLogsStream' with an 'Info' link. A 'Delete Firehose stream' button is located in the top right corner. The 'Firehose stream details' section contains the following information:

Firehose stream details			
Status ✔ Active	Destination Amazon OpenSearch Service	Data transformation Enabled	Creation time April 14, 2024 at 12:29 GMT+7
Source Direct PUT	ARN arn:aws:firehose:ap-southeast-1:654654377830:deliverystream/WebLogsStream	Error logs status ✔ 0 Destination error logs ✔ 0 Backup error logs	

At the bottom of the details section, there is a 'Test with demo data' button with an 'Info' link. The footer of the console includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

AWS Lambda

The screenshot displays the AWS Lambda console interface for a function named 'ProcessOrders'. The top navigation bar includes the AWS logo, 'Services' link, a search bar, and regional/user information (Singapore, Yunata). The breadcrumb trail shows 'Lambda > Functions > ProcessOrders'. The main header area contains the function name 'ProcessOrders' and action buttons: 'Throttle', 'Copy ARN', and 'Actions'. Below this, the 'Function overview' section is active, showing tabs for 'Diagram' and 'Template'. The 'Diagram' tab displays a visual representation of the function's configuration, including a 'ProcessOrders' function box, a 'Layers' section with '(0)' layers, and a 'Kinesis' trigger box. A '+ Add destination' button is visible next to the trigger. To the right of the diagram, a metadata panel lists: 'Description' (empty), 'Last modified' (4 hours ago), 'Function ARN' (arn:aws:lambda:ap-southeast-1:6546543778:30:function:ProcessOrders), and 'Function URL' (empty). The bottom footer contains links for 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with 'Privacy', 'Terms', and 'Cookie preferences' links.

aws Services Search [Alt+S] Singapore Yunata

Lambda > Functions > ProcessOrders

ProcessOrders

Throttle Copy ARN Actions

Function overview Info

Export to Application Composer Download

Diagram Template

ProcessOrders

Layers (0)

Kinesis

+ Add destination

Description

-

Last modified

4 hours ago

Function ARN

arn:aws:lambda:ap-southeast-1:6546543778:30:function:ProcessOrders

Function URL Info

-

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AWS Lambda

The screenshot displays the AWS Lambda console interface for a function named 'LogToJSON'. The top navigation bar includes the AWS logo, 'Services' menu, a search bar, and regional settings for 'Singapore' and 'Yunata'. The breadcrumb trail shows 'Lambda > Functions > LogToJSON'. The function title 'LogToJSON' is prominently displayed, accompanied by 'Throttle', 'Copy ARN', and 'Actions' buttons. Below this, the 'Function overview' section is active, showing tabs for 'Diagram' and 'Template'. The 'Diagram' tab displays a visual representation of the function with its icon, name 'LogToJSON', and a 'Layers' section indicating '(0)' layers. There are buttons for '+ Add trigger' and '+ Add destination'. To the right, a metadata panel lists 'Description' (empty), 'Last modified' (3 hours ago), 'Function ARN' (arn:aws:lambda:ap-southeast-1:6546543778:30:function:LogToJSON), and 'Function URL' (empty). The footer contains links for 'CloudShell', 'Feedback', and legal notices from Amazon Web Services, Inc.

aws Services Search [Alt+S] Singapore Yunata

Lambda > Functions > LogToJSON

LogToJSON

Throttle Copy ARN Actions

▼ Function overview Info

Export to Application Composer Download

Diagram Template

LogToJSON

Layers (0)

+ Add trigger + Add destination

Description -

Last modified 3 hours ago

Function ARN
arn:aws:lambda:ap-southeast-1:6546543778:30:function:LogToJSON

Function URL Info -

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AWS Glue

The screenshot displays the AWS Glue console interface. On the left is a navigation sidebar with categories like ETL jobs, Data Catalog, and Data Integration and ETL. The main content area shows the 'orderlogs' database details, including its properties and a list of tables. The 'orderlogs' database was last updated on April 14, 2024, at 08:38:38. It contains two tables: 'es_error' and 'year_2024'. The 'es_error' table is a JSON table located at 's3://orderlogs-sc', and the 'year_2024' table is a CSV table located at 's3://orderlogs-sc'. Both tables have a 'View data' link and a 'View data quality' link.

Database properties

Name	Description	Location	Created on (UTC)
orderlogs	-	-	April 14, 2024 at 06:14:48

Tables (2) Last updated (UTC) April 14, 2024 at 08:38:39

View and manage all available tables.

Filter tables

☐	Name	Database	Location	Classific...	Depreca...	View data	Data quality
☐	es_error	orderlogs	s3://orderlogs-sc	JSON	-	Table data	View data quality
☐	year_2024	orderlogs	s3://orderlogs-sc	CSV	-	Table data	View data quality

AWS Glue

The screenshot displays the AWS Glue console interface. On the left is a navigation sidebar with categories like ETL jobs, Data Catalog, and Data Integration and ETL. The main content area shows the 'year_2024' table under the 'Tables' section. It includes a breadcrumb trail 'AWS Glue > Tables > year_2024', a title 'year_2024', and metadata such as 'Last updated (UTC) April 14, 2024 at 06:22:18', a refresh button, a version dropdown set to 'Version 2 (Current version)', and an 'Actions' menu. Below this are tabs for 'Table overview' and 'Data quality New'. The 'Table details' tab is active, showing a table with four columns: Name, Description, Database, and Classification. The 'Advanced properties' tab is also visible, showing a table with three columns: Input format, Output format, and Serde serialization lib.

aws Services Search [Alt+S] Singapore Yunata

ETL jobs
Visual ETL
Notebooks
Job run monitoring
Data Catalog tables
Data connections
Workflows (orchestration)

▼ Data Catalog
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings

▼ Data Integration and ETL
ETL jobs

AWS Glue > Tables > year_2024

year_2024

Last updated (UTC) April 14, 2024 at 06:22:18 [Refresh] Version 2 (Current version) Actions

Table overview Data quality New

Table details Advanced properties

Name	Description	Database	Classification
year_2024	-	orderlogs	CSV

Location	Connection	Deprecated	Last updated
s3://orderlogs-sgp-test/year=2024/	-	-	April 14, 2024 at 06:22:18

Input format	Output format	Serde serialization lib
org.apache.hadoop.mapred.T extInputFormat	org.apache.hadoop.hive ql.io. HiveIgnoreKeyTextOutputFor mat	org.apache.hadoop.hive.serde 2.lazy.LazySimpleSerDe

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AWS Glue

The screenshot displays the AWS Glue console interface. On the left is a navigation sidebar with categories like ETL jobs, Data Catalog, and Data Integration and ETL. The main content area shows the 'es_error' table details under the 'Tables' section. The breadcrumb path is 'AWS Glue > Tables > es_error'. The table's last update is 'April 14, 2024 at 06:18:15' (UTC), and it is 'Version 1 (Current version)'. There are tabs for 'Table overview' and 'Data quality New'. The 'Table details' tab is active, showing a table with four columns: Name, Description, Database, and Classification. Below this, there are rows for Location, Input format, Output format, and Serde serialization lib.

Table details

Name	Description	Database	Classification
es_error	-	orderlogs	JSON

Location	Connection	Deprecated	Last updated
s3://orderlogs-sgp-test/es-error/	-	-	April 14, 2024 at 06:18:15

Input format	Output format	Serde serialization lib
org.apache.hadoop.mapred.T extInputFormat	org.apache.hadoop.hive.ql.io. HiveIgnoreKeyTextOutputFor mat	org.openx.data.jsonserde.Jso nSerDe

AWS Glue

The screenshot displays the AWS Glue console interface. On the left is a navigation sidebar with categories like 'Data Catalog tables', 'Data connections', 'Workflows (orchestration)', 'Data Catalog', and 'Data Integration and ETL'. The 'Crawlers' link under 'Data Catalog' is highlighted. The main content area shows the configuration for the 'order data crawler'. At the top, there's a breadcrumb trail 'AWS Glue > Crawlers > order data crawler' and a status bar indicating it was last updated on April 14, 2024, with buttons for 'Run crawler', 'Edit', and 'Delete'. Below this is a 'Crawler properties' section with a table of key-value pairs. At the bottom, there are tabs for 'Crawler runs', 'Schedule', 'Data sources', 'Classifiers', and 'Tags', with 'Crawler runs' being the active tab.

order data crawler

Last updated (UTC)
April 14, 2024 at 08:39:03

Crawler properties

Name	order data crawler	IAM role	AWSGlueServiceRole-OrderLogs	Database	orderlogs	State	READY
Description	-	Security configuration	-	Lake Formation configuration	-	Table prefix	-
Maximum table threshold	-						

► **Advanced settings**

Crawler runs | Schedule | Data sources | Classifiers | Tags

AWS Glue

The screenshot displays the AWS Glue console interface. On the left is a navigation sidebar with categories like 'Data Catalog tables', 'Data connections', 'Workflows (orchestration)', 'Data Catalog' (expanded), 'Data Integration and ETL', and 'Catalog settings'. The main content area shows the configuration for a crawler, including fields for 'Description', 'Security configuration', 'Lake Formation configuration', and 'Table prefix'. Below this is an 'Advanced settings' section. A tabbed interface at the bottom of the main area includes 'Crawler runs' (selected), 'Schedule', 'Data sources', 'Classifiers', and 'Tags'. The 'Crawler runs' tab shows a table with one entry: a completed run on April 14, 2024, at 06:17:14, lasting 01 min 01 s. The footer contains links for 'CloudShell', 'Feedback', and legal notices.

aws Services Search [Alt+S] Singapore Yunata

Data Catalog tables
Data connections
Workflows (orchestration)
▼ Data Catalog
Databases
Tables
Stream schema registries
Schemas
Connections
Crawlers
Classifiers
Catalog settings
▼ Data Integration and ETL
ETL jobs
Visual ETL
Notebooks
Job run monitoring
Interactive Sessions

Description
Security configuration
Lake Formation configuration
Table prefix
Maximum table threshold
Advanced settings

Crawler runs Schedule Data sources Classifiers Tags

Crawler runs (1)
The list of crawler runs for this crawler.
Filter data Filter by a date and time range
Start time (UTC) End time (UTC) Current/last duration Status DPU hours
April 14, 2024 at 06:17:14 April 14, 2024 at 06:18:16 01 min 01 s Completed

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Amazon DynamoDB

The screenshot displays the Amazon DynamoDB console interface. At the top, the AWS logo and 'Services' menu are visible, along with a search bar and navigation icons. The left sidebar contains a navigation menu with options like 'Dashboard', 'Tables', 'Explore items', 'PartiQL editor', 'Backups', 'Exports to S3', 'Imports from S3', 'Integrations', 'Reserved capacity', and 'Settings'. The main content area shows the 'Test-Orders' table configuration. A 'Tables (1)' filter panel is open, showing search criteria for tags and table names. The 'Test-Orders' table is selected. The 'General information' section provides details about the table's configuration, including the partition key, sort key, capacity mode, table status, and alarms.

Test-Orders

General information

Partition key CustomerID (Number)	Sort key OrderID (String)
Capacity mode Provisioned	Table status Active
Alarms No active alarms	Point-in-time recovery (PITR) Off

Amazon DynamoDB

aws Services Search [Alt+S] Singapore Yunata

Completed. Read capacity units consumed: 0.5

Items returned (26) [Refresh] [Actions] [Create item]

< 1 > [Settings] [Fullscreen]

☐	CustomerID ... ▾	OrderID (String) ▾	Country ▾	Description ▾	OrderDate ▾
☐	12748	578757-16216	United King...	LETTER SHAP...	11/25/201...
☐	12748	578757-20669	United King...	RED HEART L...	11/25/201...
☐	12748	578757-20963	United King...	APPLE BATH ...	11/25/201...
☐	12748	578757-21428	United King...	SET3 BOOK B...	11/25/201...
☐	12748	578757-21429	United King...	RED GINGHA...	11/25/201...
☐	12748	578757-21710	United King...	FOLDING UM...	11/25/201...
☐	12748	578757-22142	United King...	CHRISTMAS C...	11/25/201...

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Amazon Athena

The screenshot displays the Amazon Athena Query Editor interface. At the top, the AWS logo and 'Services' menu are visible, along with a search bar and a keyboard shortcut '[Alt+S]'. The main header shows 'Amazon Athena > Query editor'. Below this, there are tabs for 'Editor', 'Recent queries', 'Saved queries', and 'Settings'. A 'Workgroup' dropdown menu is set to 'primary'. A notification banner states: 'Athena now supports typeahead code suggestions to speed up SQL query development. Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.' The left sidebar contains a 'Data' section with 'Data source' (set to 'AwsDataCatalog') and 'Database' (set to 'orderlogs'). Below this is a 'Tables and views' section with a 'Create' button. The main area shows a query editor with a single query labeled 'Query 1' containing the following SQL code:

```
1 select description, count(1) as order_count from year_2024 where country = 'France' group by description
```

. The bottom of the interface includes a footer with 'CloudShell', 'Feedback', and copyright information: '© 2024, Amazon Web Services, Inc. or its affiliates.' along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

Amazon Athena

The screenshot displays the Amazon Athena console interface. At the top, the AWS logo and 'Services' menu are visible, along with a search bar and navigation icons. The main interface is divided into a left sidebar and a central workspace.

Left Sidebar:

- Data:** Includes a refresh icon and a back arrow.
- Data source:** A dropdown menu set to 'AwsDataCatalog'.
- Database:** A dropdown menu set to 'orderlogs'.
- Tables and views:** Includes a 'Create' button and a search bar labeled 'Filter tables and views'.
- Tables (2):** A list of tables: 'es_error' (Partitioned) and 'year_2024' (Partitioned).
- Views (0):** A section for views, currently empty.

Central Workspace:

- Query 1:** A SQL query is entered: `select description, count(1) as order_count from year_2024 where country = 'France' group by description`.
- SQL Editor:** The query is displayed in a dark-themed editor with line numbers.
- Execution Controls:** At the bottom of the workspace, there are buttons for 'Run again' (highlighted in orange), 'Explain', 'Cancel', 'Clear', and 'Create'. To the right of these buttons is a toggle switch for 'Reuse query results up to 60 minutes ago'.

Footer:

- CloudShell icon and 'Feedback' link.
- Copyright notice: '© 2024, Amazon Web Services, Inc. or its affiliates.'
- Links for 'Privacy', 'Terms', and 'Cookie preferences'.

AWS Billing and Cost Management (Budgets)

The screenshot displays the AWS Billing and Cost Management console interface. The left sidebar contains navigation links for Billing and Cost Management, Home, Getting Started, Billing and Payments, Bills, Payments, Credits, Purchase Orders, Cost Analysis, Cost Explorer, Cost Explorer Saved Reports, Cost Anomaly Detection, and Free Tier. The main content area shows the 'Data-Engineer-Budget' details, including a 'Budget health' section with 'Current vs. budgeted' and 'Forecasted vs budgeted (MTD)' metrics, both at 0.00%. The 'Alerts' section shows 'Thresholds' as 'OK' and 'Actions' as '-'. The 'Details' section at the bottom lists 'Budget name', 'Budget amount', and 'Start date'. The top navigation bar includes the AWS logo, Services, Search, and various utility icons. The bottom footer contains CloudShell, Feedback, and copyright information.

Billing and Cost Management

Home [New](#)

Getting Started [New](#)

Billing and Payments

Bills

Payments

Credits

Purchase Orders

Cost Analysis

Cost Explorer [New](#)

Cost Explorer Saved Reports

Cost Anomaly Detection

Free Tier

Billing and Cost Management > Budgets > Budget details

Data-Engineer-Budget [Info](#)

[Delete](#) [Edit](#)

Budget health [Info](#)

Current vs. budgeted	Forecasted vs budgeted (MTD)
0.00%	0.00%
Amount spent: \$0.00 of \$10.00	Amount spent: - of \$10.00

Alerts [Info](#)

Thresholds

✓ OK

Actions

-

[View all alerts](#)

Details

Budget name	Budget amount	Start date
-------------	---------------	------------

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AWS Billing and Cost Management (Budgets)

The screenshot displays the AWS Billing and Cost Management console. The left sidebar contains navigation links for Billing and Cost Management, Home, Getting Started, Billing and Payments, Bills, Payments, Credits, Purchase Orders, Cost Analysis, Cost Explorer, Cost Explorer Saved Reports, Cost Anomaly Detection, and Free Tier. The main content area shows the details of a budget named 'Data-Engineer-Budget' with a budget amount of \$10.00 and a start date of 2024-04-01. The budget type is 'Cost budget' and the period is 'Monthly'. A button labeled 'View all alerts' is visible in the top right. The bottom of the console shows the 'Budget history' tab and a 'View in AWS Cost Explorer' button.

Billing and Cost Management

Home [New](#)

Getting Started [New](#)

Billing and Payments

Bills

Payments

Credits

Purchase Orders

Cost Analysis

Cost Explorer [New](#)

Cost Explorer Saved Reports

Cost Anomaly Detection

Free Tier

Details

Budget name	Budget amount	Start date
Data-Engineer-Budget	\$10.00	2024-04-01
Budget type	Period	End date
Cost budget Info	Monthly	-

[▶ Additional budget parameters](#)

[Budget history](#) [Alerts](#)

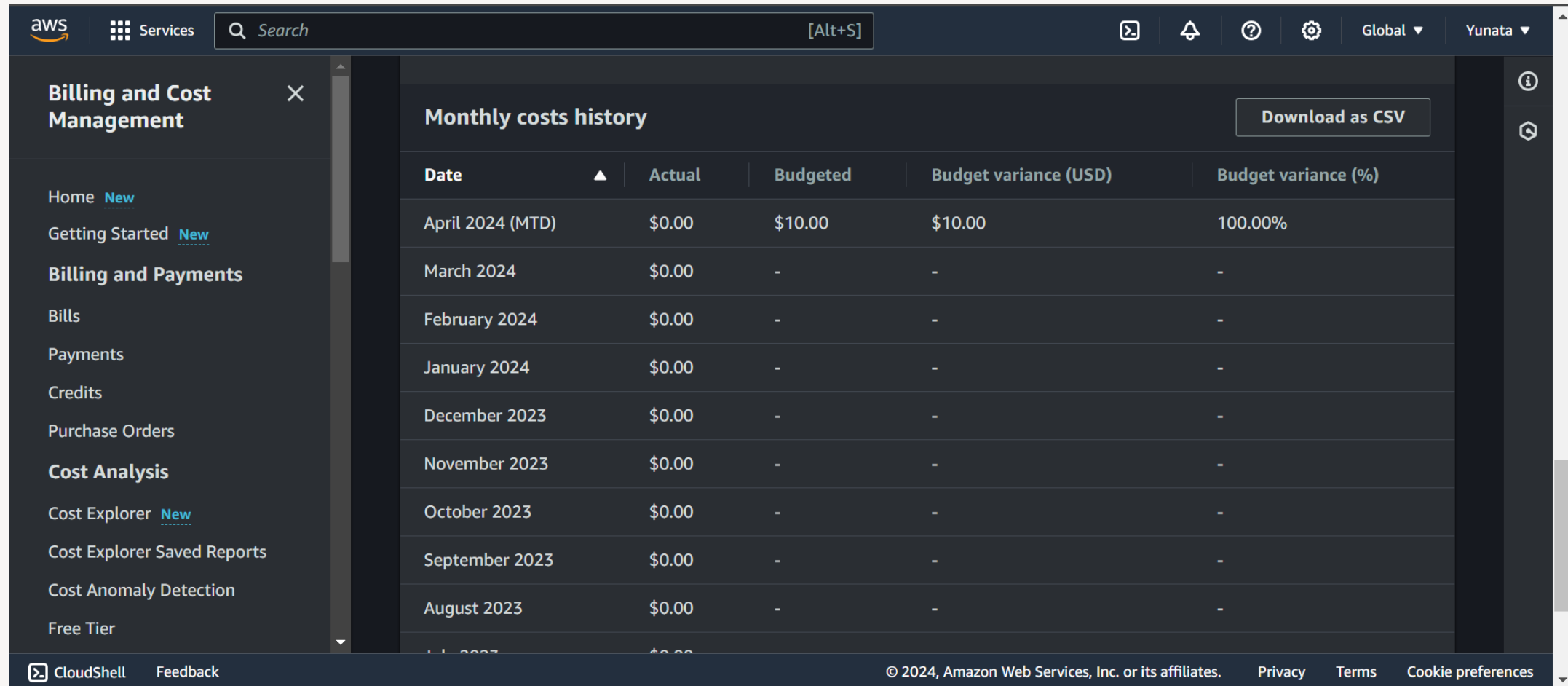
Budget history

[View in AWS Cost Explorer](#)

CloudShell Feedback

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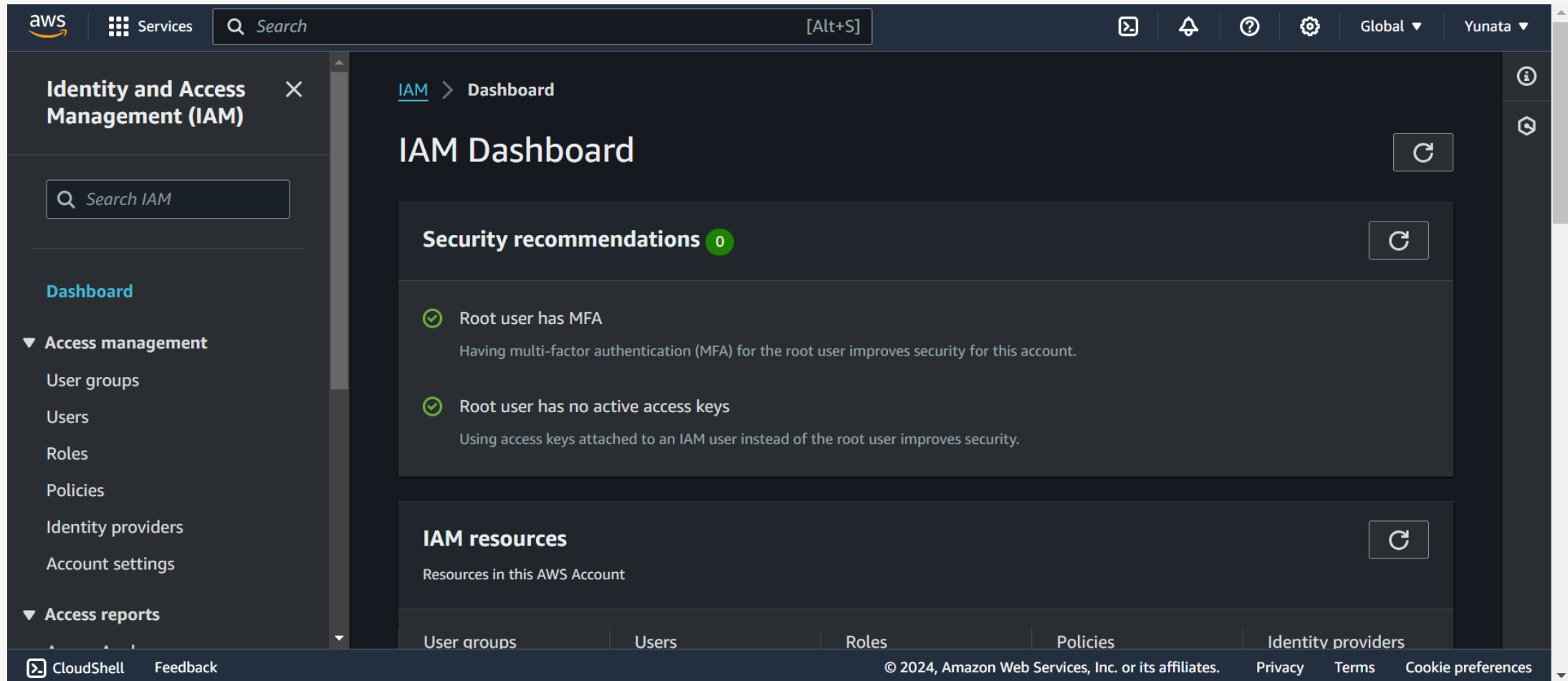
AWS Billing and Cost Management (Budgets)



The screenshot displays the AWS Billing and Cost Management console. The left sidebar contains navigation links for Home, Getting Started, Billing and Payments, Cost Analysis, and Free Tier. The main content area shows a table titled 'Monthly costs history' with columns for Date, Actual, Budgeted, Budget variance (USD), and Budget variance (%). The table lists data for April 2024 (MTD) and various months from March 2024 back to July 2023. A 'Download as CSV' button is located in the top right corner of the table area. The footer includes links for CloudShell, Feedback, and legal notices.

Monthly costs history					Download as CSV
Date	Actual	Budgeted	Budget variance (USD)	Budget variance (%)	
April 2024 (MTD)	\$0.00	\$10.00	\$10.00	100.00%	
March 2024	\$0.00	-	-	-	
February 2024	\$0.00	-	-	-	
January 2024	\$0.00	-	-	-	
December 2023	\$0.00	-	-	-	
November 2023	\$0.00	-	-	-	
October 2023	\$0.00	-	-	-	
September 2023	\$0.00	-	-	-	
August 2023	\$0.00	-	-	-	
July 2023	\$0.00	-	-	-	

Amazon Identity and Access Management



Amazon IAM Users

The screenshot shows the Amazon IAM console interface. The top navigation bar includes the AWS logo, 'Services' menu, a search bar, and utility icons. The left sidebar is titled 'Identity and Access Management (IAM)' and contains a search bar and a list of navigation items: Dashboard, Access management (expanded), Users (selected), Roles, Policies, Identity providers, Account settings, and Access reports. The main content area is titled 'IAM > Users' and shows 'Users (1)' with an 'Info' link. Below this is a description: 'An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.' There is a search bar and a table of users. The table has columns: User name, Path, Group, Last activity, MFA, and Password status. One user is listed: 'EC2-UserAdmin' with a path of '/' and no group. At the bottom of the console, there is a footer with 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates, along with links for Privacy, Terms, and Cookie preferences.

☐	User name	Path	Group	Last activity	MFA	Password
☐	EC2-UserAdmin	/	-	-	-	-

Amazon IAM Users

Access key created
This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

[IAM](#) > [Users](#) > [EC2-UserAdmin](#) > Create access key

Step 1
[Access key best practices & alternatives](#)

Step 2 - optional
[Set description tag](#)

Step 3
Retrieve access keys

Retrieve access keys [Info](#)

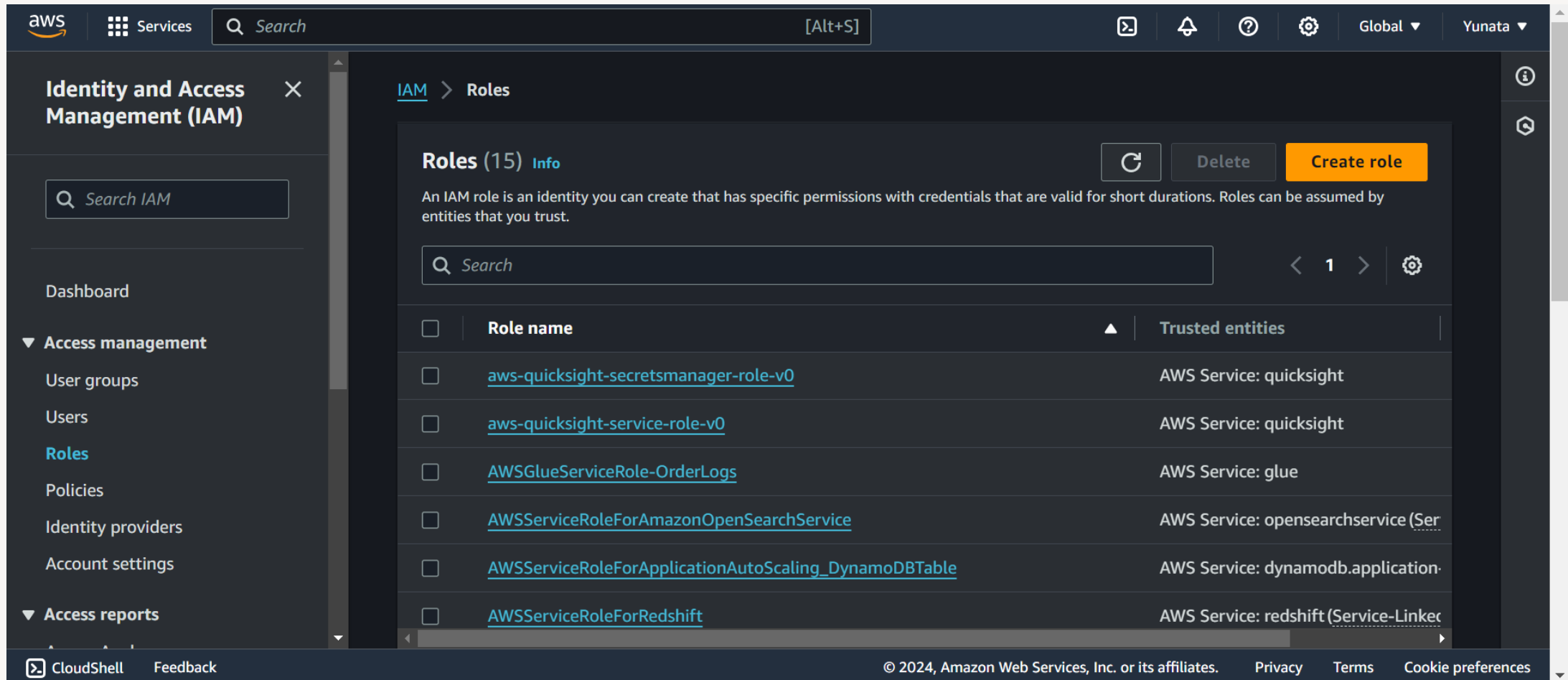
Access key
If you lose or forget your secret access key, you cannot retrieve it. Instead, create a new access key and make the old key inactive.

Access key	Secret access key
 [Redacted]	 [Redacted] Hide

Access key best practices

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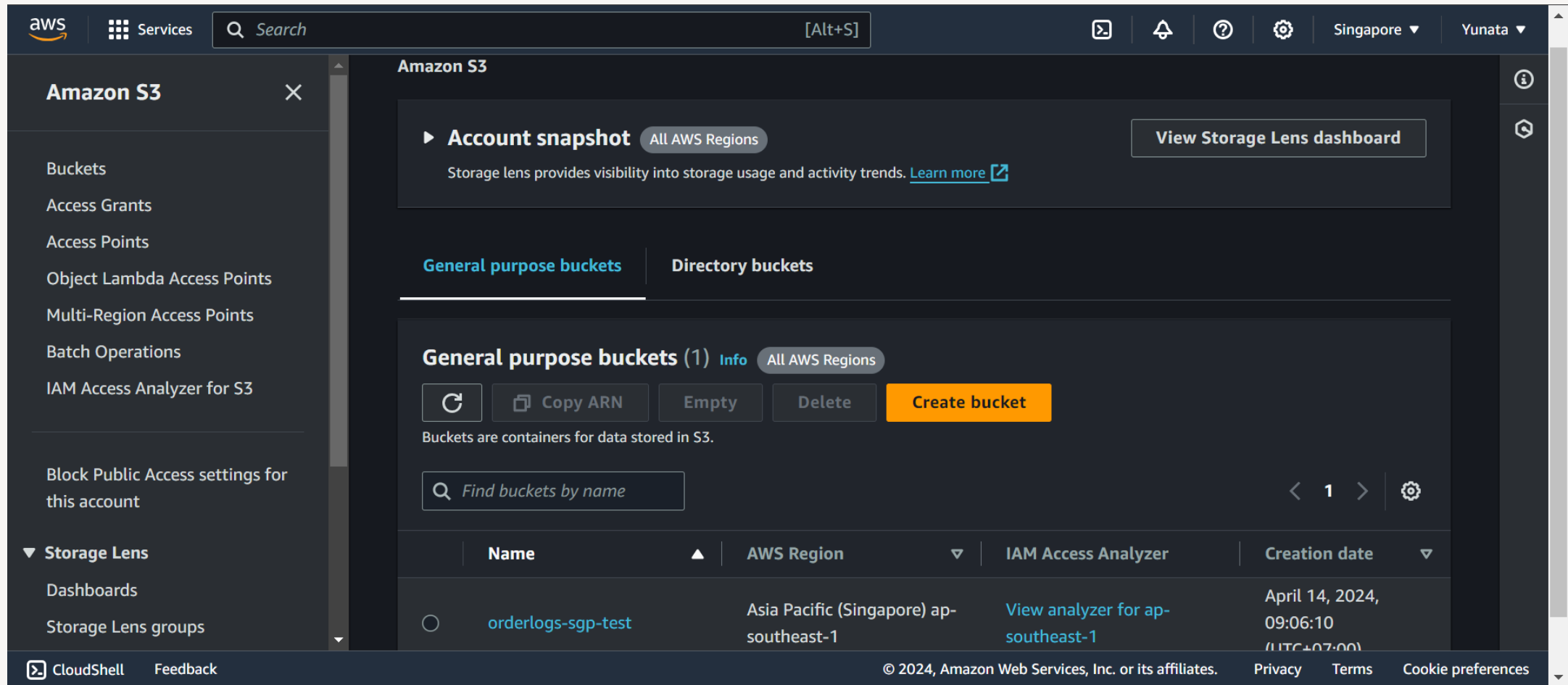
Amazon IAM Roles



The screenshot displays the Amazon IAM console interface. The left sidebar shows the navigation menu with 'Identity and Access Management (IAM)' selected. The main content area is titled 'IAM > Roles' and shows 'Roles (15)' with an 'Info' link. A description states: 'An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.' Below this is a search bar and a table of roles.

☐	Role name	Trusted entities
☐	aws-quicksight-secretsmanager-role-v0	AWS Service: quicksight
☐	aws-quicksight-service-role-v0	AWS Service: quicksight
☐	AWSGlueServiceRole-OrderLogs	AWS Service: glue
☐	AWSServiceRoleForAmazonOpenSearchService	AWS Service: opensearchservice (Ser
☐	AWSServiceRoleForApplicationAutoScaling_DynamoDBTable	AWS Service: dynamodb.application
☐	AWSServiceRoleForRedshift	AWS Service: redshift (Service-Linker

Amazon Simple Storage Service (S3)



The screenshot displays the Amazon S3 console interface. The top navigation bar includes the AWS logo, a search bar, and regional settings for Singapore. The left sidebar lists various S3 services, with 'Amazon S3' selected. The main content area is titled 'Amazon S3' and features an 'Account snapshot' section with a 'View Storage Lens dashboard' button. Below this, there are tabs for 'General purpose buckets' and 'Directory buckets'. The 'General purpose buckets' tab is active, showing a list of buckets. A table at the bottom lists the buckets, with one bucket named 'orderlogs-sgp-test' visible. The table columns are Name, AWS Region, IAM Access Analyzer, and Creation date.

Name	AWS Region	IAM Access Analyzer	Creation date
orderlogs-sgp-test	Asia Pacific (Singapore) ap-southeast-1	View analyzer for ap-southeast-1	April 14, 2024, 09:06:10 (UTC+07:00)

Amazon Simple Storage Service (S3)

The screenshot displays the Amazon S3 console interface. The top navigation bar includes the AWS logo, a 'Services' menu, a search bar, and regional/user information (Singapore, Yunata). The left sidebar shows the 'Amazon S3' section with various options like Buckets, Access Grants, and Storage Lens. The main content area is titled 'Objects (3)' and includes tabs for Objects, Properties, Permissions, Metrics, Management, and Access Points. Below the tabs, there are action buttons: Refresh, Copy S3 URI, Copy URL, Download, Open, Delete, and an Actions dropdown. A 'Create folder' button and a highlighted 'Upload' button are also present. A descriptive text block explains that objects are fundamental entities in S3 and provides a link to 'Amazon S3 inventory'. Below this is a search bar 'Find objects by prefix' and a table listing three folders: 'athena/', 'es-error/', and 'year=2024/'. The footer contains links for CloudShell, Feedback, and copyright information.

Amazon S3

Services Search [Alt+S]

Singapore Yunata

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

Dashboards

Storage Lens groups

Objects (3) Info

Refresh Copy S3 URI Copy URL Download Open Delete Actions

Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	athena/	Folder	-	-	-
☐	es-error/	Folder	-	-	-
☐	year=2024/	Folder	-	-	-

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Update v2 Confluent + AWS

Solved:

- Same Region
- IAM Policy

Cluster Setting

The screenshot displays the Confluent Cloud user interface. At the top, a header bar shows a credit balance of \$390/\$400, a 'Free trial' status with a 'credit left' indicator, and an 'Enter payment' button. A notification states, 'You are currently using a free trial version of Confluent Cloud'. The main navigation bar includes the Confluent logo, a search bar for 'Stream Catalog', and links for 'LEARN', notifications, help, and a menu. The left sidebar lists navigation options: 'Data portal', 'Stream processing' (marked 'New'), 'Cluster links', 'Stream shares', and 'Support'. The central content area shows the 'cluster_0' settings page. It features a cluster status card with a 'Running' indicator and a three-dot menu. Below this, the 'Metrics' section displays 'Production' at 1B/s, 'Consumption' at 3B/s, and 'Storage' at 86.52KB. The 'Resources' section shows 'ksqlDB' at 1, 'Connectors' at 3, and 'Clients' at 23. The 'Overview' section lists the cluster ID as 'lkc-wj8wqw', Type as 'Basic', and Provider & region as 'AWS | ap-southeast-1'. The right sidebar, titled 'Show more metadata', includes a 'Stream Governance package' section with an 'Upgrade now' button, and a 'Stream Governance API' section with a chat icon. The 'Schemas' link is highlighted in the right sidebar.

Cluster Settings for cluster_0

Metrics

Production	Consumption	Storage
1B/s	3B/s	86.52KB

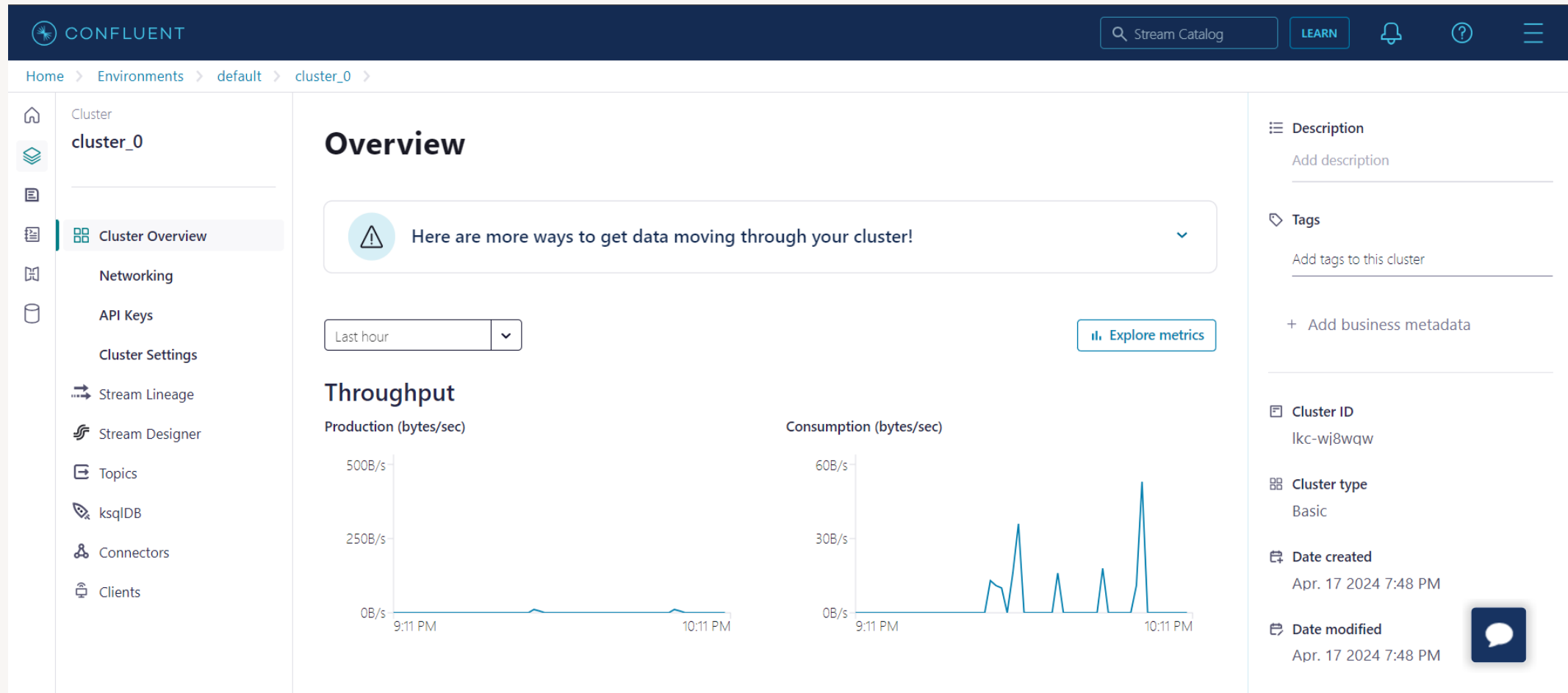
Resources

ksqlDB	Connectors	Clients
1	3	23

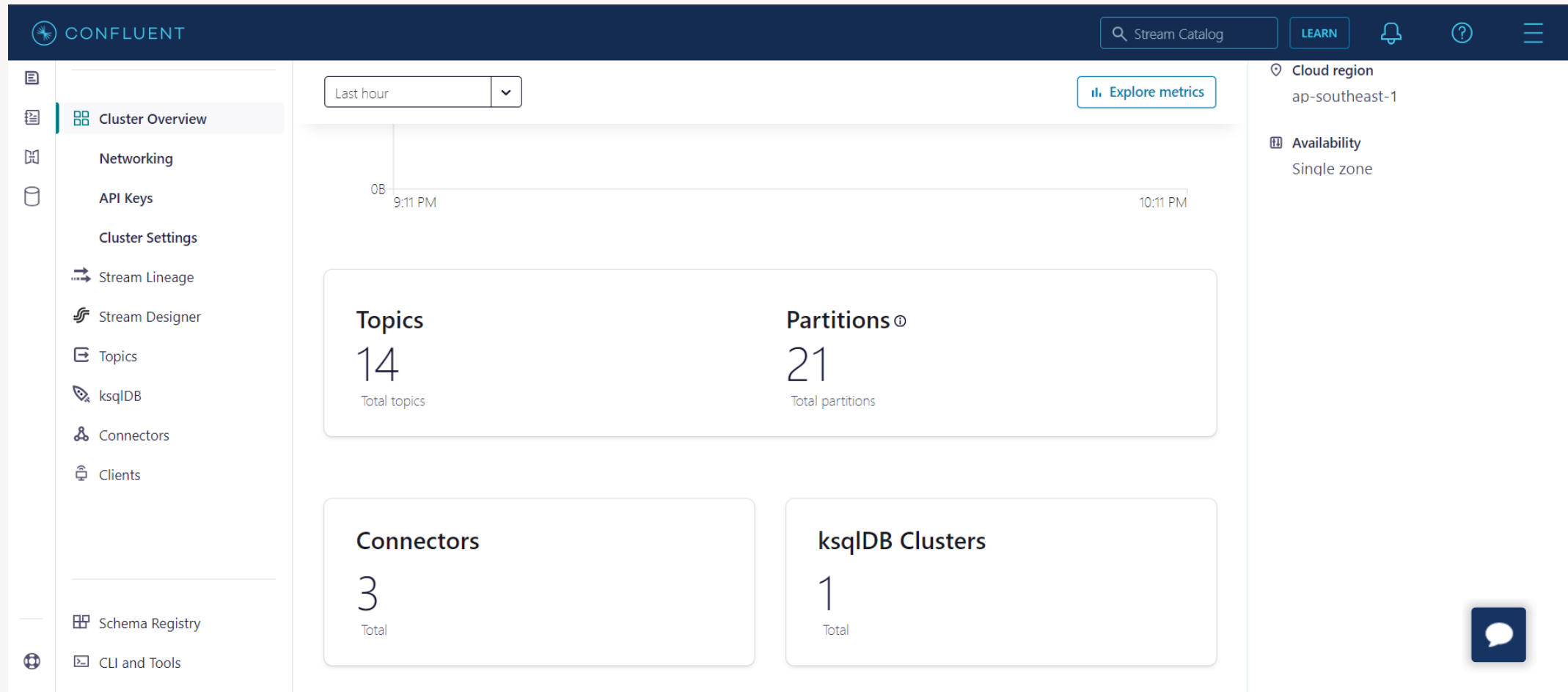
Overview

ID	lkc-wj8wqw
Type	Basic
Provider & region	AWS ap-southeast-1

Cluster Overview



Cluster Overview



Connectors

The screenshot shows the Confluent Cloud console interface. The top navigation bar includes the Confluent logo, a search bar for the Stream Catalog, a 'LEARN' button, and notification and help icons. The breadcrumb trail indicates the current location: Home > Environments > default > cluster_0 > Connectors >. The left sidebar contains a navigation menu with options like Cluster Overview, Networking, API Keys, Cluster Settings, Stream Lineage, Stream Designer, Topics, ksqlDB, Connectors (which is highlighted), and Clients. The main content area is titled 'Connectors' and features a search bar. It displays three connector cards, each with a status indicator (Running) and a table of metrics. Below these cards is a section for 'Connect with popular connectors' showing options like Amazon S3 Sink, Snowflake Sink, Elasticsearch Service Sink, and MongoDB Atlas Source. A '+ Add Connector' button is located in the top right corner of the main content area.

Connectors

Search by connector or plugin name

DynamoDb_Sink_Connector_0 Running

Tasks	Bytes/sec
1	0B/s
Messages/sec	Messages behind
0	0

Overview

Category	Sink
ID	lcc-95no97
Plugin name	Amazon DynamoDb Sink

S3_SINK_Connector_0 Running

Tasks	Bytes/sec
1	0B/s
Messages/sec	Messages behind
0	1

Overview

Category	Sink
ID	lcc-nrgpjd
Plugin name	Amazon S3 Sink

S3_SINK_Connector_1 Running

Tasks	Bytes/sec
1	--
Messages/sec	Messages behind
--	1

Overview

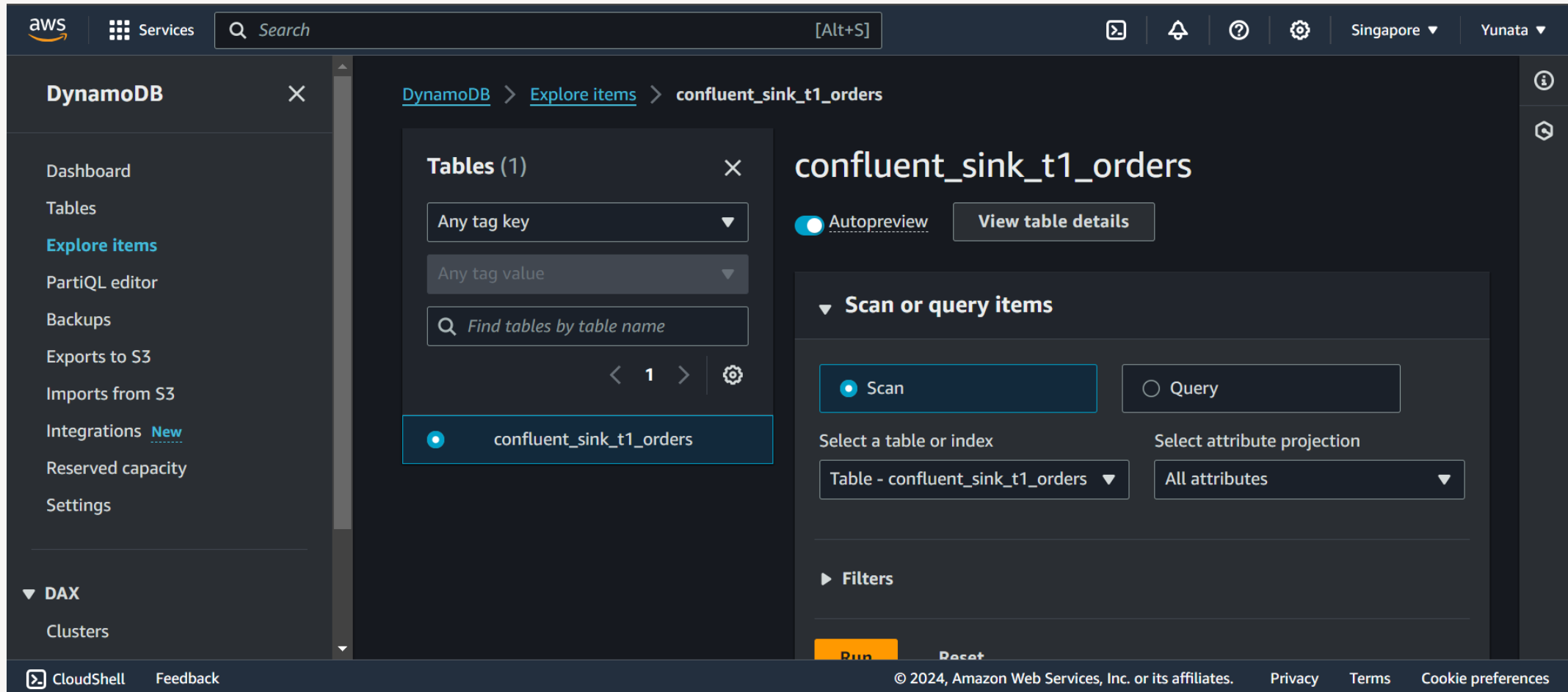
Category	Sink
ID	lcc-wj80gw
Plugin name	Amazon S3 Sink

Connect with popular connectors

- Amazon S3 Sink
- Snowflake Sink
- Elasticsearch Service Sink
- MongoDB Atlas Source

See all connectors

Amazon DynamoDB Sink Table



Amazon DynamoDB Sink Table

The screenshot displays the AWS Management Console interface for the Amazon DynamoDB Sink Table. The top navigation bar includes the AWS logo, a search bar, and regional settings for Singapore. The left sidebar lists various DynamoDB services, with 'DynamoDB' selected. The main content area shows the 'Sink Table' configuration page. A green notification bar indicates a successful completion: 'Completed. Read capacity units consumed: 0.5'. Below this, the 'Items returned (1)' section shows a table with one item. The table has columns for 'partition (Number)', 'offset (Number)', and 'AGE'. The first item has a partition of 0, an offset of 7, and an AGE of 29. The bottom of the console shows the footer with copyright information and links to Privacy, Terms, and Cookie preferences.

partition (Number)	offset (Number)	AGE
0	7	29

Amazon S3 Data Lake Bucket (New)

aws

Services

Search

[Alt+S]

🔍

🔔

?

⚙️

Singapore

Yunata

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

Amazon S3

Account snapshot

All AWS Regions

View Storage Lens dashboard

Storage lens provides visibility into storage usage and activity trends. [Learn more](#)

General purpose buckets

Directory buckets

General purpose buckets (1) [Info](#)

All AWS Regions

🔄

📄 Copy ARN

Empty

Delete

Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

< 1 > ⚙️

Name	AWS Region	IAM Access Analyzer	Creation date
s3s-dea1	Asia Pacific (Singapore) ap-southeast-1	View analyzer for ap-southeast-1	April 17, 2024, 19:20:50 (UTC+07:00)

CloudShell

Feedback

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Modern Data Platform Using Confluent and AWS

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Amazon S3 (Connector Generate Directory)

The screenshot displays the Amazon S3 console interface. The left sidebar contains navigation options: Buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, IAM Access Analyzer for S3, Block Public Access settings for this account, Storage Lens (expanded), Dashboards, Storage Lens groups, and AWS Organizations settings. The main content area shows the 'topics/' directory. The 'Objects (2)' section lists two folders: 'sink_t1_orders/' and 'src-api-orders/'. The table below shows the details of these objects.

	Name	Type	Last modified	Size	Storage class
☐	📁 sink_t1_orders/	Folder	-	-	-
☐	📁 src-api-orders/	Folder	-	-	-

Amazon S3 “sink_t1_orders” Sink (AVRO)

The screenshot displays the Amazon S3 console interface. The left sidebar shows the navigation menu with options like Buckets, Access Grants, and Storage Lens. The main content area shows the path: Amazon S3 > Buckets > s3s-dea1 > topics/ > sink_t1_orders/ > year=2024/ > month=04/ > day=17/ > hour=14/. The 'hour=14/' folder is selected, and the 'Objects' tab is active. A table lists one object: 'sink_t1_orders+0+000000006.avro' of type 'avro', last modified on April 17, 2024, with a size of 1.1 KB and a storage class of 'Standard'. The bottom of the console shows the footer with copyright information and links to Privacy, Terms, and Cookie preferences.

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

CloudShell Feedback

Amazon S3 > Buckets > s3s-dea1 > topics/ > sink_t1_orders/ > year=2024/ > month=04/ > day=17/ > hour=14/

hour=14/

Copy S3 URI

Objects (1) Info

Upload

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
☐	sink_t1_orders+0+000000006.avro	avro	April 17, 2024, 22:01:28 (UTC+07:00)	1.1 KB	Standard

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Amazon S3 “src-api-orders” Sink (JSON)

The screenshot displays the Amazon S3 console interface. The left sidebar shows the 'Amazon S3' service with various navigation options like Buckets, Access Grants, and Storage Lens. The main content area shows the path: `Amazon S3 > Buckets > s3s-dea1 > topics/ > src-api-orders/ > year=2024/ > month=04/ > day=17/ > hour=14/`. The current view is 'hour=14/'. Below the path, there are tabs for 'Objects' and 'Properties'. The 'Objects' tab is active, showing a list of objects. There is one object listed: `src-api-orders+0+00000000 04.json`, which is a JSON file of size 122.0 B, created on April 17, 2024, at 22:01:28 (UTC+07:00), and stored in the Standard storage class. The console also includes a search bar, a 'Copy S3 URI' button, and a 'Create folder' button. The bottom of the console shows the 'CloudShell' button and 'Feedback' link.

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Amazon S3 > Buckets > s3s-dea1 > topics/ > src-api-orders/ > year=2024/ > month=04/ > day=17/ > hour=14/

hour=14/

Copy S3 URI

Objects (1) Info

Refresh Copy S3 URI Copy URL Download Open Delete Actions Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	<code>src-api-orders+0+00000000 04.json</code>	json	April 17, 2024, 22:01:28 (UTC+07:00)	122.0 B	Standard

CloudShell Feedback

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Amazon IAM Policy (External Service)

aws

Services

Search

[Alt+S]

🔍

🔔

?

⚙️

Global

Yunata

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies (SCPs)

Related consoles

CloudShellFeedback

IAM > Policies > Confluent-Policy

Confluent-Policy

EditDelete

Policy details

Type	Creation time	Edited time	ARN
Customer managed	April 14, 2024, 20:13 (UTC+07:00)	April 17, 2024, 21:59 (UTC+07:00)	arn:aws:iam::654654377830:policy/Confluent-Policy

Permissions

Entities attached

Tags

Policy versions (4)

Access Advisor

This policy defines some actions, resources, or conditions that do not provide permissions. To grant access, policies must have an action that has an applicable resource or condition. For details, choose **Show remaining**. [Learn more](#)

Permissions defined in this policy

EditSummaryJSON

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it

Search

Allow (2 of 408 services)

Show remaining 406 services

Service	Access level	Resource	Request condition
DynamoDB	Limited: Read, Write	All resources	None
S3	Limited: Read, List, Tagging, Write	Multiple	None

Modern Data Platform Using Confluent and AWS

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JSON Policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListAllMyBuckets"
      ],
      "Resource": "arn:aws:s3:*:*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetBucketLocation",
        "s3:ListBucketMultipartUploads"
      ],
      "Resource": "arn:aws:s3:::s3s-dea1"
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:PutObjectTagging",
        "s3:GetObject",
        "s3:AbortMultipartUpload",
        "s3:ListMultipartUploadParts"
      ],
      "Resource": "arn:aws:s3:::s3s-dea1/*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "dynamodb:CreateTable",
        "dynamodb:BatchWriteItem",
        "dynamodb:Scan",
        "dynamodb:DescribeTable"
      ],
      "Resource": "*"
    }
  ]
}
```

Data JSON

[Items]

```
{
  "itemid": 1,
  "fruit": "Apple",
  "size": "Large",
  "color": "Red"
}
{
  "itemid": 2,
  "fruit": "Banana",
  "size": "Medium",
  "color": "Yellow"
}
{
  "itemid": 3,
  "fruit": "Lime",
  "size": "Small",
  "color": "Green"
}
```

[Users]

```
{
  "userid": 1,
  "name": "Mary Jane",
  "age": 29,
  "height": 164,
  "weight": 50
}
{
  "userid": 2,
  "name": "Glen",
  "age": 21,
  "height": 175,
  "weight": 65
}
```

[Orders]

```
{
  "ordertime": 1497014222380,
  "orderid": 1,
  "itemid": 1,
  "userid": 1,
  "address": {
    "city": "Mountain View",
    "state": "CA",
    "zipcode": 94041
  }
}
{
  "ordertime": 1497016222380,
  "orderid": 2,
  "itemid": 2,
  "userid": 1,
  "address": {
    "city": "Lake View",
    "state": "LV",
    "zipcode": 94044
  }
}
```

Data JSON

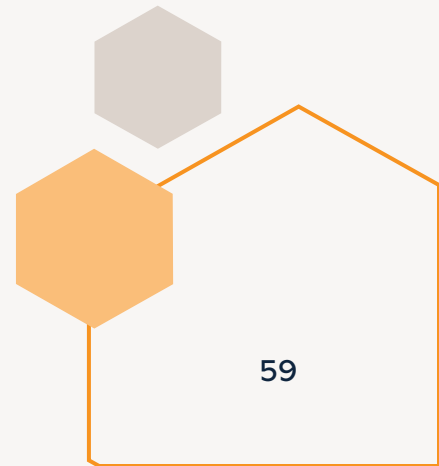
[Orders Continued]

```
{
  "ordertime": 1713364568299,
  "orderid": 5,
  "itemid": 2,
  "userid": 2,
  "address": {
    "city": "Melbourne",
    "state": "ME",
    "zipcode": 90000
  }
}

{
  "ordertime": 1713664568299,
  "orderid": 6,
  "itemid": 3,
  "userid": 1,
  "address": {
    "city": "Melbourne",
    "state": "ME",
    "zipcode": 90000
  }
}
```

```
{
  "ordertime": 1497020222349,
  "orderid": 3,
  "itemid": 3,
  "userid": 2,
  "address": {
    "city": "Los Angeles",
    "state": "LA",
    "zipcode": 93047
  }
}

{
  "ordertime": 1497020222349,
  "orderid": 4,
  "itemid": 1,
  "userid": 2,
  "address": {
    "city": "Melbourne",
    "state": "ME",
    "zipcode": 90000
  }
}
```



ksqlDB Query (Notes)

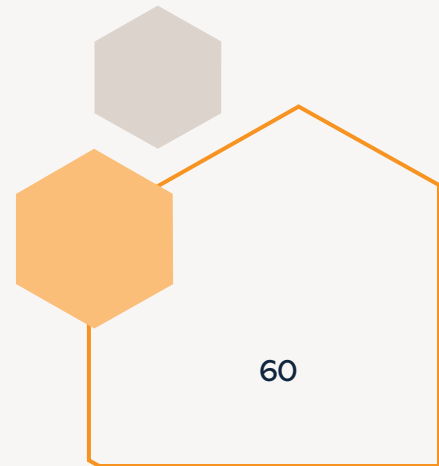
Note:

- "TABLE" directly from topic doesn't need aggregation function, "TABLE" from "STREAM" require aggregation function
- for some reason, integer primary key on json serdes is not working if is printed on string (probably wrong serdes????)

```
CREATE TABLE tblO_api_items (itemid INTEGER PRIMARY KEY, fruit STRING, `size` STRING, color STRING)
    WITH (kafka_topic='src-api-items', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON');
CREATE TABLE tblO_api_users (userid INTEGER PRIMARY KEY, name STRING, age INTEGER, height INTEGER, weight INTEGER)
    WITH (kafka_topic='src-api-users', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON');
```

Note:

- varchar primary key on json serde is working
 - beware creating table directly on topic is not materialized, if on stream or another table it is materialized
- ```
CREATE TABLE tblO_api_items_v5 (itemid VARCHAR PRIMARY KEY, fruit STRING, `size` STRING, color STRING)
 WITH (kafka_topic='src-api-items', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON', PARTITIONS=1);
CREATE TABLE tblO_api_users_v5 (userid VARCHAR PRIMARY KEY, name STRING, age INTEGER, height INTEGER, weight INTEGER)
 WITH (kafka_topic='src-api-users', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON', PARTITIONS=1);
```

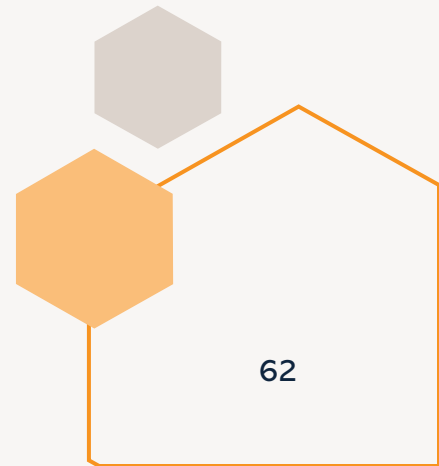


# ksqlDB Query (Part 1)

```
CREATE STREAM strO_api_items (itemid INTEGER, fruit STRING, `size` STRING, color STRING)
 WITH (kafka_topic='src-api-items', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON');
CREATE STREAM strO_api_users (userid INTEGER, name STRING, age INTEGER, height INTEGER, weight INTEGER)
 WITH (kafka_topic='src-api-users', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON');
CREATE STREAM strO_api_orders (ordertime TIMESTAMP, orderid INTEGER, itemid INTEGER, userid INTEGER, address STRUCT<city STRING, state STRING, zipcode
INTEGER>)
 WITH (kafka_topic='src-api-orders', KEY_FORMAT='KAFKA', VALUE_FORMAT='JSON', PARTITIONS=1);
```

# ksqlDB Query (Part 2)

```
CREATE OR REPLACE STREAM str1_orders WITH (KAFKA_TOPIC='sink_t1_orders', PARTITIONS=1, REPLICAS=3, KEY_FORMAT='AVRO', VALUE_FORMAT='AVRO') AS
SELECT
 CONCAT(CAST(A.ORDERID AS STRING), B.ITEMID, C.USERID) ROWKEY,
 AS_VALUE(CONCAT(CAST(A.ORDERID AS STRING), B.ITEMID, C.USERID)) UNIQUE_KEY_ID,
 A.ORDERTIME ORDERTIME,
 A.ORDERID ORDERID,
 CAST(B.ITEMID AS INTEGER) ITEMID,
 CAST(C.USERID AS INTEGER) USERID,
 A.ADDRESS->CITY CITY,
 A.ADDRESS->STATE STATE,
 A.ADDRESS->ZIPCODE ZIPCODE,
 B.FRUIT FRUIT,
 B.`size` FRUITSIZE,
 B.COLOR COLOR,
 C.NAME NAME,
 C.AGE AGE,
 C.HEIGHT HEIGHT,
 C.WEIGHT WEIGHT
FROM strO_api_orders A
JOIN tblO_api_items_v5 B ON CAST(A.ITEMID AS STRING) = B.ITEMID
JOIN tblO_api_users_v5 C ON CAST(A.USERID AS STRING) = C.USERID
WHERE A.ORDERID IS NOT NULL AND B.ITEMID IS NOT NULL AND C.USERID IS NOT NULL
PARTITION BY CONCAT(CAST(A.ORDERID AS STRING), B.ITEMID, C.USERID) EMIT CHANGES;
```



**Thank you**

